**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Programming: Arrays and Iteration Practice

**Name** _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

## Do Now - Four retrieval questions

1. What makes a system embedded?

2. How many input combinations do three Boolean inputs have?
   A) 3
   B) 6
   C) 9
   D) 8

3. "Invalid test data must always have the wrong data type." Correct this. Explain why.

4. Draw a structure diagram: Quiz contains Menu, Play and Results. Play contains Input and Scoring. Why is this not a flowchart of running steps?

## Today's objectives

Explain the core ideas of programming: arrays and iteration practice using accurate OCR terminology. Apply a clear method to the worked and guided questions.

## Key vocabulary - Read, cover, recall

**Array iteration**
A loop visits elements by their valid indexes.

**Accumulator**
A running total starts at zero and adds each element.

**Counting condition**
A counter increments only when a tested item matches a rule.

**Off-by-one error**
Accessing index length or missing final element changes the result or causes a fault.

**BLACK PEN: Write one definition in your own words.**

| Do Now | I Do | We Do | You Do | Review & Improve |
|---|---|---|---|---|
| **Recall previous learning** | Read new knowledge | Practise with guidance | Work independently | Check my understanding |

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# Learn and practise

## New Learning - Key ideas

**Array iteration**
A loop visits elements by their valid indexes.

**Accumulator**
A running total starts at zero and adds each element.

**Counting condition**
A counter increments only when a tested item matches a rule.

**Off-by-one error**
Accessing index length or missing final element changes the result or causes a fault.

## I Do - Worked model

**Calculate the total of [3,5,2,4] using a running total.**

- Start total=0.
- After values 3 and 5, total=8.
- Add 2 to reach 10, then 4 to reach 14.
- Print 14; the four valid indexes are 0-3.

## We Do - Guided practice

1. Count values greater than 5 in [2,7,9,4].

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**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Think and write

## Code First - Predict > Write > Test > Improve

**YOUR TASK: Write the missing statement that adds the current file size.**

```
array filesMB = [3,5,2]
total = 0
for i = 0 to 2
...
next i
print(total)
```

## You Do - Independent exam questions

Q2. Give the output of summing [4,1,3]. [2 marks]

Q3. Explain why index 4 is invalid for a four-element zero-indexed array. [2 marks]

Q4. Explain how to count even values in an array. [3 marks]

## Exit Ticket - True or False?

A three-element zero-indexed array has last index 3.

**TRUE [ ]** **FALSE [ ]**

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Review & improve

GREEN PEN: Tick each correct point; correct mistakes.

## Do Now - Model answers

1. It is built into a larger device for a specific task, often with limited resources.

2. D - 8

3. 12 is an integer but outside the valid range, so it must be rejected.

4. Quiz branches into Menu, Play and Results. Play branches into Input and Scoring. This shows parts and their hierarchy, rather than the order they run.

## I Do - Explain the model

Start total=0.

After values 3 and 5, total=8.

Add 2 to reach 10, then 4 to reach 14.

Print 14; the four valid indexes are 0-3.

## We Do - Model answers

1. Final count=2.

## Code First - Answer

```
array filesMB = [3,5,2]
total = 0
for i = 0 to 2
total = total + filesMB[i]
next i
print(total)
```

Expected output: 10

## You Do - Self-mark

Q2: 8.

Q3: Valid indexes are 0,1,2,3; index 4 is out of bounds.

Q4: Iterate each element; if value MOD 2 = 0 then increment counter; output counter.

**GREEN PEN: Choose the hardest answer. Correct it on lined paper.**